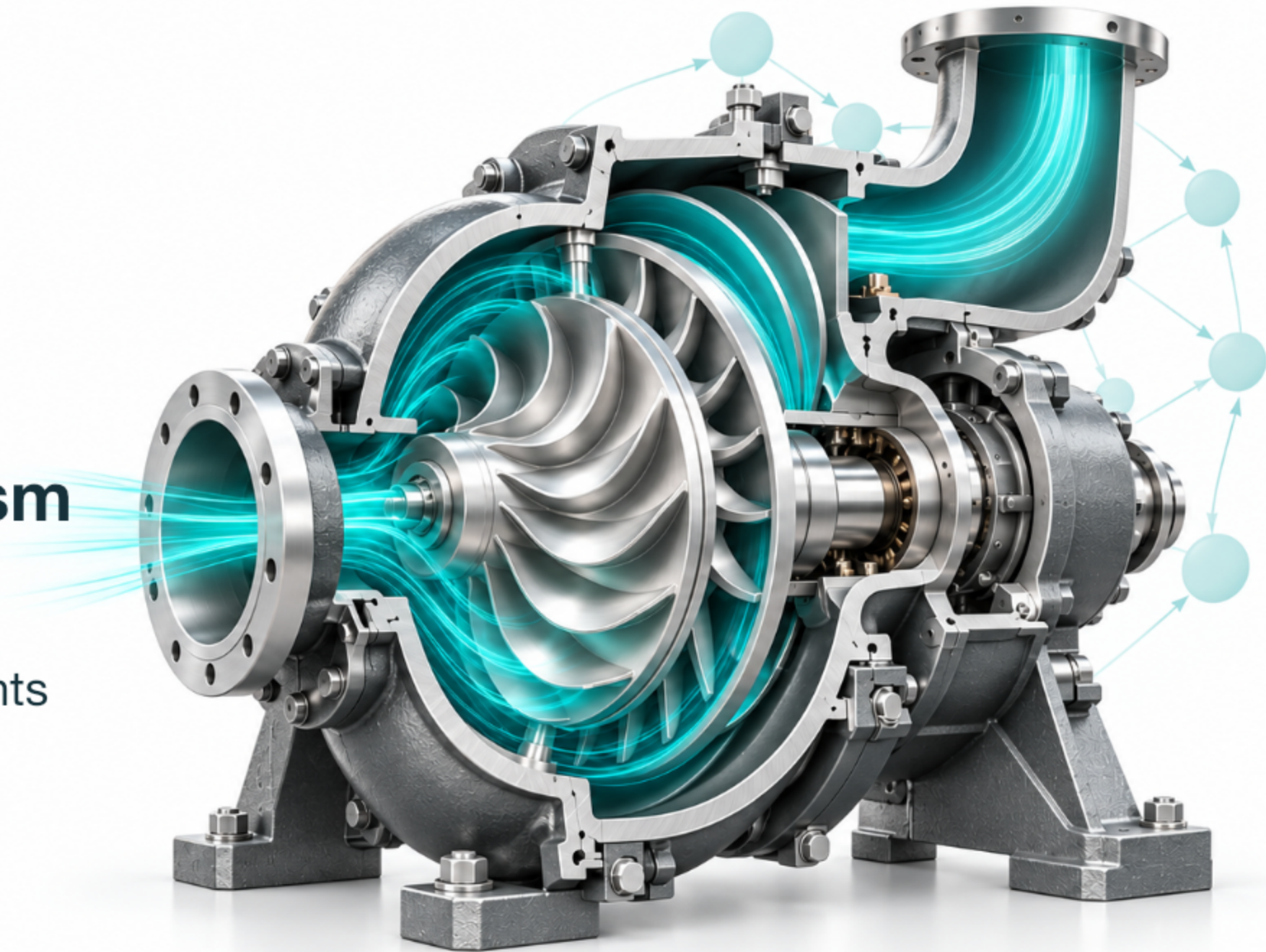


ACE

Active Causal Experimentalism

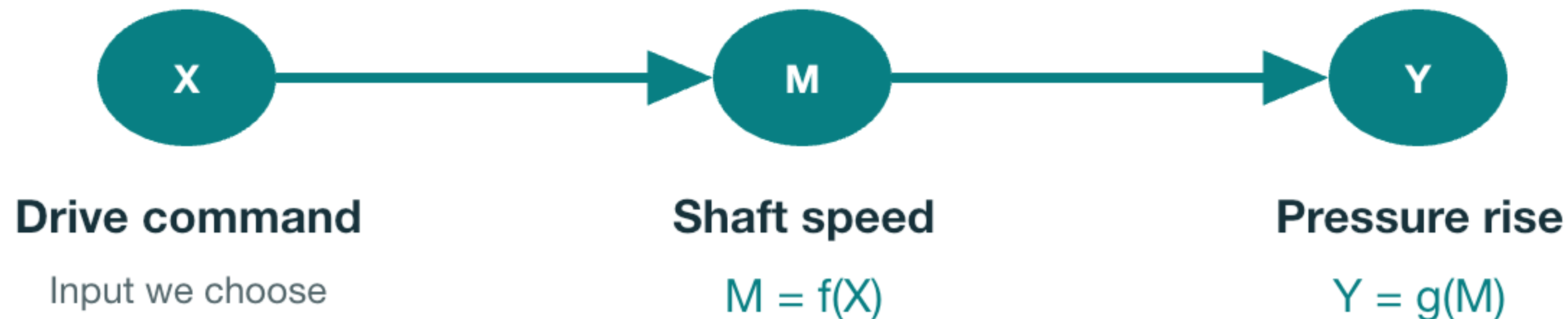
Every experiment counts



Compressor concept illustration

A compressor as a causal model

Illustrative test rig with other operating conditions fixed



An SCM models each local mechanism, then connects its predictions.

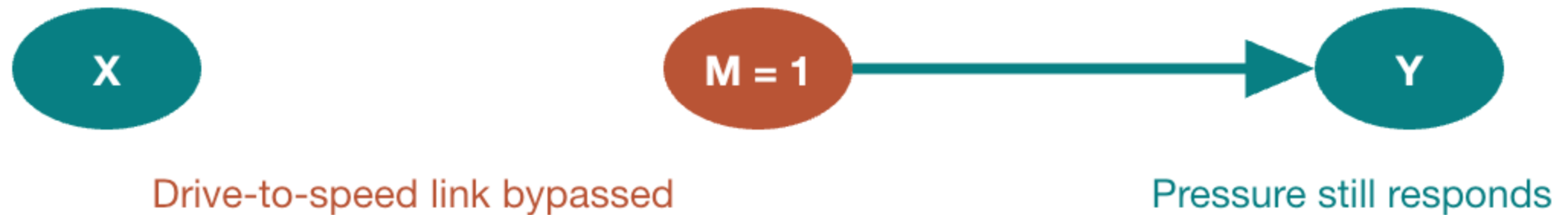
Teaching values: $M = 2.5X$, $Y = 3M$. Normalized deviations around a reference condition.

Two interventions on the same compressor

Change the drive command



Hold shaft speed with a test-rig controller: $\text{do}(M = 1)$

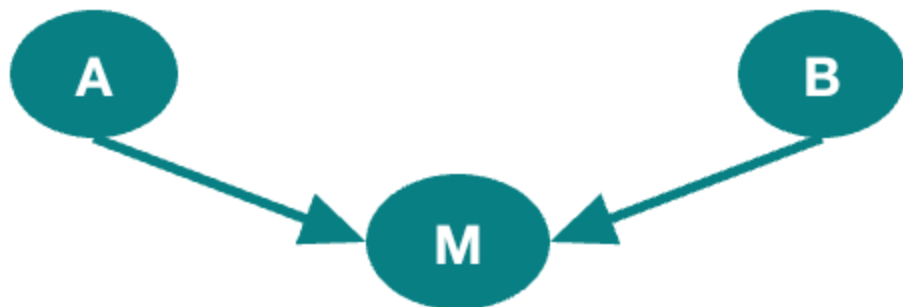


Learn only from mechanisms left natural. The imposed speed is not a training label for $f(X)$.

Local mechanism tables vs one joint table

Same system: 10 inputs, 5 settings each, 9 mechanisms with two parents

SCM LOCAL TABLES



A	B	$M = f(A, B)$
0	0	m_1
0	1	m_2
...	...	...
4	4	m_{25}

25 rows per mechanism $\times$ 9 = 225

Discrete illustration. Local maps compose through the graph. ACE fits functions instead of literal tables.

SCM-FREE JOINT TABLE

All input settings together determine the final output Y

X_1	X_2	...	X_{10}	Y
0	0	...	0	y_1
0	0	...	1	y_2
...	...	...	...	...
4	4	...	4	y_n

$5^{10} = 9,765,625$ joint rows

Exhaustive grids multiply the work

10 inputs · 5 settings each · a deliberately naive joint-table baseline

SCM-FREE JOINT TABLE

9.77 million

configurations for exhaustive coverage

29.26 million random draws

for 95% expected joint-grid coverage¹

SCM LOCAL TABLES

225

local entries

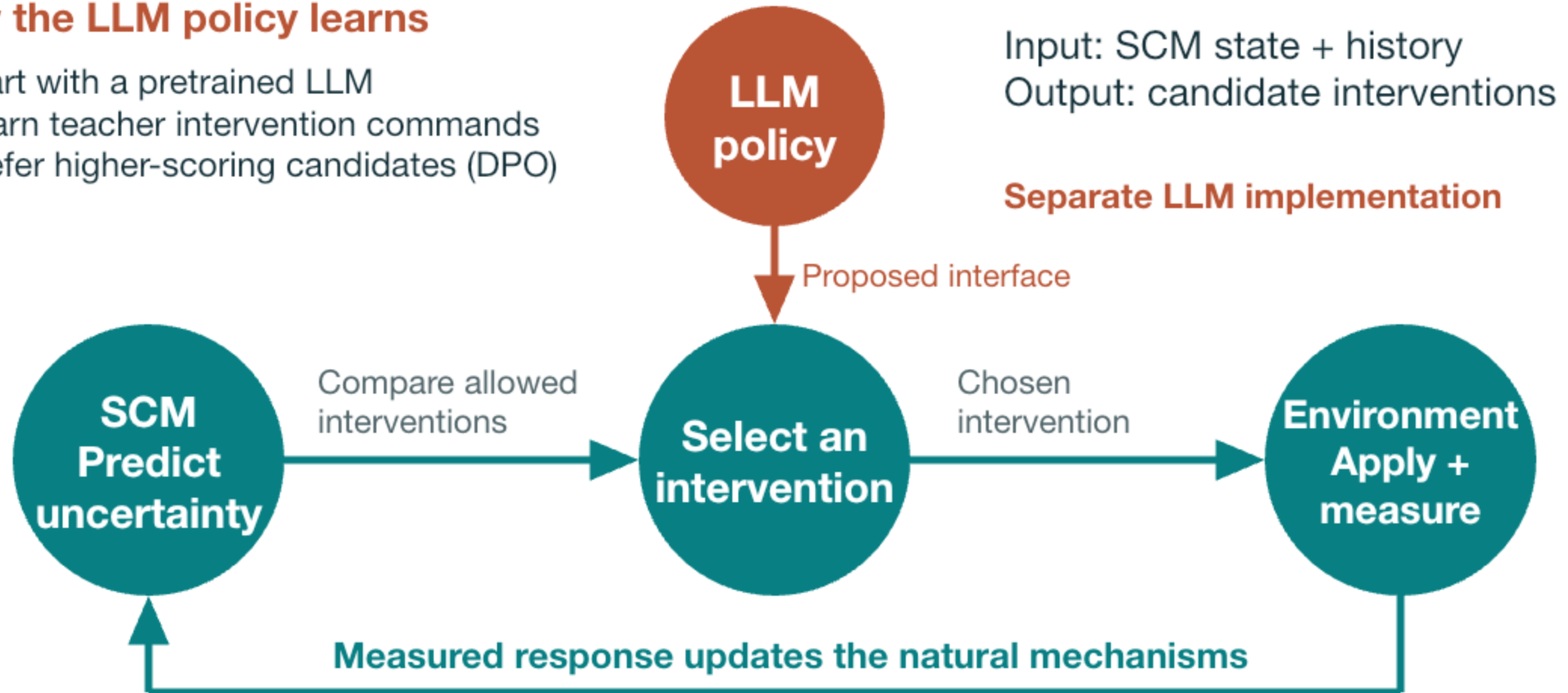
9 mechanisms
25 settings each

Analytic coverage / representation example, not a measured ACE speedup. ¹ Uniform draws with replacement.

The ACE intervention loop

How the LLM policy learns

1. Start with a pretrained LLM
2. Learn teacher intervention commands
3. Prefer higher-scoring candidates (DPO)



Allowed interventions: drive X or speed M at -1 , 0 , $+1$. Stop when the budget ends.

The plotted results use direct SCM scoring (PEV), without the LLM policy.

Exact inputs for the compressor example

Mechanism	Member slopes	Mean prediction
M from X	1, 2, 3	$\hat{M} = 2X$
Y from M	2, 3, 4	$\hat{Y} = 3M$

Interventions: drive X or shaft speed M at -1, 0 or +1

References: -1, 0, +1 Noise proxy: 0.05

Budget: one response Ties: first intervention in the list

Normalized deviations: low -1, reference 0, high +1. Y is pressure rise.

One intervention can inform two mechanisms

Intervention	M contribution	Y contribution	Total
do(X = -1)	0.413437	0.436264	0.849701
do(X = 0)	0.000000	0.000000	0.000000
do(X = 1)	0.413437	0.436264	0.849701
do(M = -1)	0.000000	0.413437	0.413437
do(M = 0)	0.000000	0.000000	0.000000
do(M = 1)	0.000000	0.413437	0.413437

Choose drive X = -1. Forecast: $\bar{M} = -2$, $\hat{Y} = -6$

Score contribution: $\text{mean Cov}^2 / (\text{Variance} + \text{noise proxy}) \cdot \text{tied interventions follow the declared order}$

One measured response improves both predictions

Observed row: $X = -1$, $M = -2.5$, $Y = -7.5$

Natural mechanism	Training input	Training label
M	Measured $X = -1$	$M = -2.5$
Y	Measured $M = -2.5$	$Y = -7.5$

Learn drive $\rightarrow$ speed and measured speed $\rightarrow$ pressure

$M = -2.25$ $\hat{Y} = -6.75$ Budget left: 0

Both mechanisms stayed natural, so the same intervention response teaches both.

A recorded synthetic system: 30 mechanisms



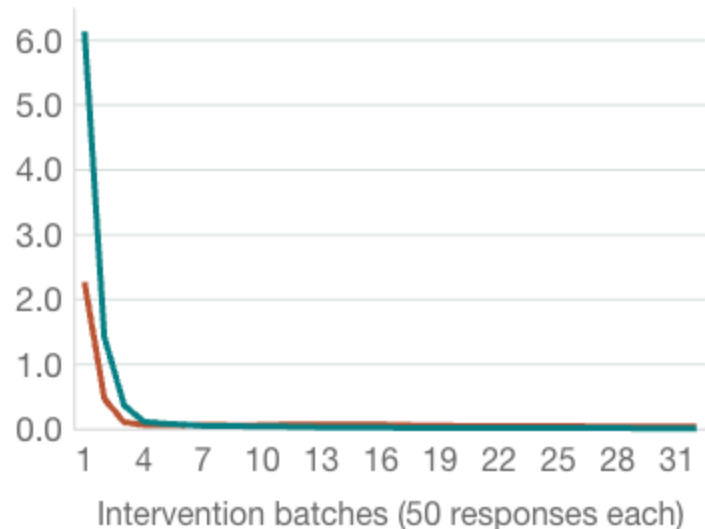
First selected intervention: $\text{do}(X7 = 3.8571)$ → 50 responses

Arrows point from cause to effect. Coral follows the X7 intervention downstream · archived seed 5000.

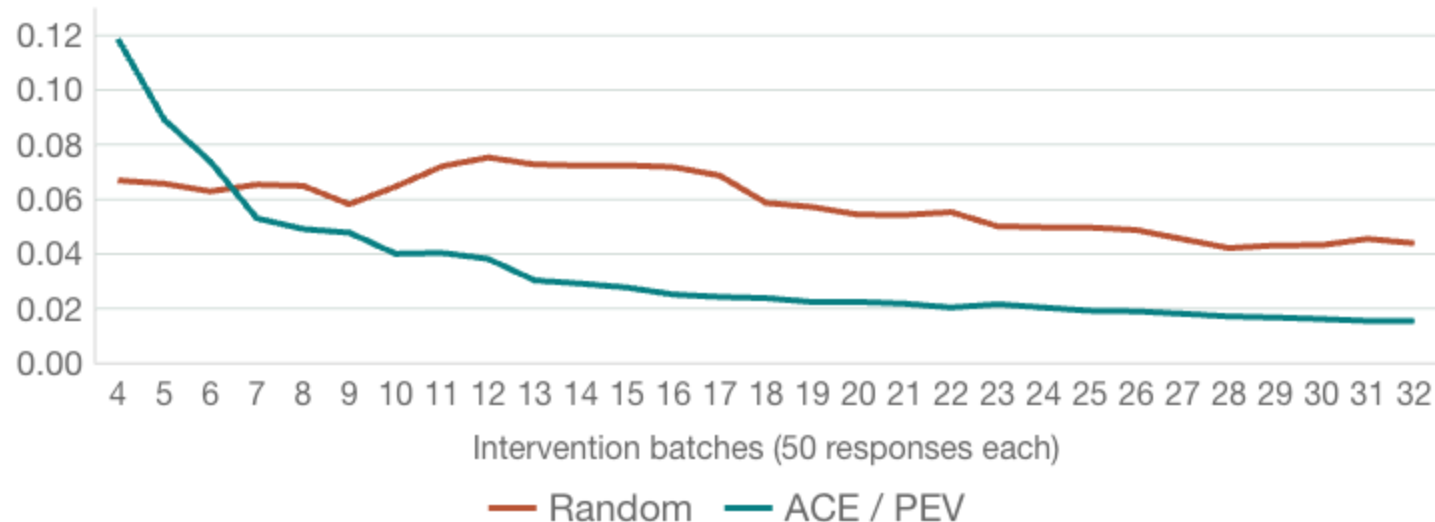
65% lower final prediction error

ACE wins on 19 of 20 systems at equal budgets

Full recorded curve



Detail: batches 4–32



Final MSE: 0.01543 ACE vs 0.04390 random

20 synthetic systems. 2,000 responses per arm. Both use the same SCM learner.

64% fewer batches in the measured comparison

Two measured intervention policies alongside an analytic grid-coverage example

Recorded: matched prediction error

Analytic: 95% grid coverage

Counts of queries (linear scale)



500 vs 1,400 intervention responses

29.26 million grid draws

First two: 10 vs 28 batches at $MSE \leq 0.04390$. Third: a different task, not a measured ACE speedup.